ANDRY PAEZ

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EXPERIENCE

Research Assistant, Dept. of Physics and Astronomy **Feb 2026 – Present**

San Jose State University (advisor: Dr. Hilary Hurst), 30 hrs/week

- Develop physics simulation software (Python/NumPy); 150+ commits merged into the research group's codebase through advisor-reviewed pull requests with reproducible uv-managed environments.
- Vectorized a core numerical operator for up to 1,880x speedup with bit-exact parity verified by a benchmark harness; built a CLI + HDF5 pipeline with a hash-addressed run catalog (SQLite) and Slurm HPC workflows.

PROJECTS

Reachy Mini Robot Control Console **Jun 2026 – Present**

- Built a FastAPI + JavaScript web control console for a Pollen Robotics Reachy Mini on a headless Jetson Orin Nano, architected as a pure HTTP/WebSocket proxy to the robot daemon that never grabs the robot lock; 162 tests passing, live-verified against hardware on the Jetson.
- Fixed a motor-safety race with a mode-epoch guard and an event-loop blocking bug; reduced idle UI overhead to zero DOM writes per tick, resolving a measured phone battery/heat problem verified by driving the served app with Playwright.
- Brought up the robot's camera and audio stack on arm64: compiled GStreamer webrtcsink from source with NVMM buffer transport, root-caused speaker silence to ALSA card remapping and PipeWire sink volume after a JetPack 7.2/Ubuntu 24.04 migration, diagnosed XVF3800 DSP firmware read-back, and fixed a SIGINT-forwarding bug with a process-tree signaller.
- Built a voice-control app on the same platform: microphone to faster-whisper to LLM (with live camera frames for visual questions) to TTS; 286 tests passing.

omnisense, Offline Voice and Vision Assistant for Jetson **Apr 2026**

- Built a modular edge-perception package for the Jetson Orin Nano: camera abstraction (GStreamer/OpenCV/mock backends), faster-whisper speech-to-text, voice activity detection, YOLO object-detection overlay, and a FastAPI REST/WebSocket/MJPEG interface; mock backends make the perception stack testable without hardware.

ESP32 Embedded Firmware **Apr 2026**

- esp32-jetson-stats: ESP32 + ST7796 display firmware (FreeRTOS dual-core, PlatformIO/C++) rendering live Jetson CPU/GPU/MEM/TEMP telemetry streamed over serial JSON (about 861 LOC plus a Python sender).
- esp32-jetson-launcher: touchscreen launcher (PlatformIO C++, TFT_eSPI/FT6X36) with the Jetson streaming RGB565 framebuffer stripes over WiFi.

OPEN SOURCE

- **NousResearch/hermes-agent:** two merged bugfixes to a large open-source agent framework — a systemd gateway restart race and a Discord 8KB command-payload split — real async/systems debugging.

SKILLS

- **Robotics and Embedded:** Jetson Orin Nano (JetPack, 24/7 edge inference server), ESP32 firmware (FreeRTOS, PlatformIO, C++), serial protocols, real-time event loops, motor-safety guards, hardware bring-up
- **Perception and Media:** camera pipelines (GStreamer/NVMM, OpenCV), YOLO object detection, faster-whisper STT, ALSA/PipeWire audio
- **Languages and Systems:** Python, C/C++, Rust, TypeScript/JavaScript; FastAPI, WebSockets/SSE, Linux (systemd), Docker, pytest, Playwright-driven runtime verification, Git and PR-based review workflow

EDUCATION

B.S., Physics **May 2026**

San Jose State University. GPA 3.80. Dean's Scholar (Fall 2024, Fall 2025)

A.S.-T Computer Science, Mathematics, Physics; A.A. Language and Rationality **May 2024**

Modesto Junior College. GPA 3.73. With Honors; President's List (5 terms)